

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2025

Course: B. Tech

Branch: Electrical / Electrical Engineering(Electrical and Power)

Subject Code & Name: BTEEC601_Y19 Control System Engineering

Semester : VI

Max Marks: 60

Date:30/ 7/ 2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level /CO)	Mark
Q. 1	Objective type questions. (Compulsory Question)		12
1	The use of signal flow graph is to find which of the following a. Transfer function of control system b. Poles of control system c. Controllability of control system d. Stability of control system	CO1	1
2	Identify the correct for the transfer function of positive feedback a. $\frac{G(s)}{1-G(s)H(s)}$ b. $\frac{G(s)}{1+G(s)H(s)}$ c. $\frac{G(s)}{(s)H(s)}$ d. $\frac{G(s)}{G(s)H(s)-1}$	CO1	1
3	The steady state error is determined as the difference between the reference input and system output at a. $t=t_p$ b. $t=\infty$ c. $t=0$ d. $t=$ time constant	CO3	1
4	The steady state error is specified in terms of a. Damping factor b. Bandwidth c. Error constants d. Speed of response	CO3	1
5	For the system $\frac{C(s)}{R(s)} = \frac{16}{s^2+8s+16}$ The nature of the time response will be a. Overdamped b. underdamped c. critically damped d. None of these	CO2	1
6	The roots of the characteristic equation are same as a. closed-loop zeros b. closed-loop poles c. open-loop zeros d. open-loop poles	CO2	1
7	The Bode plot is obtained using	CO4	1

	a. characteristics equation	b. open loop transfer function	c. closed loop transfer function	d. overall transfer function		
8	A second order control system with $\zeta = 0$ is always				CO4	1
	a. marginally stable	b. stable	c. unstable	d. inherently stable		
9	The Routh-Hurwitz criterion gives				CO3	1
	a. relative stability	b. absolute stability	c. gain margin	d. phase margin		
10	The analysis of multiple-input multiple-output system is conveniently studied by				CO6	1
	a. state space approach	b. root locus approach	c. characteristic equation approach	d. Nichols chart		
11	The state transition matrix $\phi(t)$ is given by				CO6	1
	a. $[sI] - [A]$	b. $\{[sI]-[A]\}^{-1}$	c. $L^{-1}\{[sI]-[A]\}^{-1}$	d. $L^{-1}\{[sI]-[A]\}$		
12	The integral controller enhances the				CO5	1
	a. Dynamic response	b. transient response	c. steady-state response	d. No effect of the controller		
Q. 2	Solve the following.					12
A)	The transfer function of a system is given below $G(s) = \frac{8(s+3)(s+4)}{s(s+2)^2 s^2+2s+5}$ Determine the poles and zeros and show pole-zero configuration in s-plane					6
B)	Compare open loop system with feedback control system.					6
Q.3	Solve the following.					12
A)	Find the transfer function of the system shown in figure by using block diagram reduction rules.					6

B)	Explain signal flow graph with the help of Masons gain formula.		6
Q. 4	Solve Any Two of the following.		12
A)	Determine the stability of a system having following characteristic equation $s^6+s^5+5s^4+3s^3+2s^2-4s-8$		6
B)	For the system having closed loop transfer function $\frac{C(s)}{R(s)} = \frac{18}{s^2 + 4s + 18}$ Determine i) Damped frequency of oscillations ii) Peak time iii) % peak overshoot		6
C)	The overall transfer function of a unity feedback control system is given by $\frac{C(s)}{R(s)} = \frac{10}{s^2 + 6s + 10}$ Determine i) step, ramp, parabolic coefficients ii) Steady state error if input is $r(t) = 1+t+t^2$		6
Q.5	Solve Any Two of the following.		12
A)	Plot the root locus of unity feedback control system has open loop transfer function $G(s) = \frac{K}{s(s+4)}$		6
B)	Enlist the different controllers. Write short note on PID controller.		6
C)	Describe the steps to construct Bode plot for any given system.		6
Q. 6	Solve Any Two of the following.		12
A)	Find state transition matrix for $\begin{bmatrix} 0 & -2 \\ 1 & -3 \end{bmatrix}$		6
B)	Explain the term Controllability and Observability with necessary equations.		6
C)	Obtain the state model for the system with transfer function $\frac{C(s)}{R(s)} = \frac{s^2 + 3s + 2}{s^3 + 9s^2 + 26s + 24}$		6
	*** End ***		